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Article

Time-Varying Relationships Between the "Sentiment" of U.S. Public Opinion and News Media

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Abstract

Pollsters and journalists in the United States often make generalized statements about the state of the country, in particular, its "mood" or "sentiment." The claims often rely on responses to survey questions asking about the state of the country. Little is known about the ebb and flow of this sentiment, however. This paper develops a measure of public sentiment, pooling results of thousands of polls between 2000 and 2021. It then develops an analogous measure of media sentiment based on a content analysis of millions of network television news transcripts. Analyses explore differences across data sources (i.e., polling houses and television networks) and consider the relationship between public and media sentiment. Results suggest that the measures are correlated concurrently, and there is some evidence that public sentiment responds to earlier media coverage. However, connections between the measures wax and wane over time, highlighting the potential importance of accounting for time-varying parameters in the study of public opinion and media effects.

Both popular and academic accounts of U.S. public affairs regularly reference public "mood" or "sentiment." Indeed, some of the longest-standing questions in survey research on public opinion reflect an attempt to capture a kind of generalized public sentiment by asking about satisfaction with how "things are going" in the country or whether the country is headed in the "right direction." Journalists often refer to the country's positive or negative mood leading into wars or out of an economic recession, and analysts examine public sentiment as a potential indicator of which party is likely to win the presidency (Cook, 2020). References to public mood were readily evident in the news coverage in the run-up to the 2024 U.S. national election (e.g., Fox News Polling Unit, 2024; Vick, 2022)—an election that one *Axios* journalist referred to as the "mood-swing election" (Basu, 2024).

Public mood has received and continues to receive a good amount of attention, then. Despite a tendency to treat it as a meaningful quantity, there have been limited efforts to understand this kind of generalized sentiment.[1] Gallup, a prolific user of the right direction question, treats sentiment fairly abstractly, linking it to general trends in partisan enthusiasm and beliefs about problems facing the country (e.g., Saad, 2025). Indeed, generality across domains and issues is central to how pollsters and journalists use public sentiment. During the 2024 election, *The Hill* described it as a broad indicator of the public's "collective mood and potential desire for political change or stability" (Right/Wrong Track Polls, 2024). Supporting this intuition of generality, public sentiment is empirically related to a range of material factors, including presidential approval, attitudes about congress, and economic conditions; a former president of the Roper Center even speculated that the diverse measures typically linked to public sentiment "all tap a diffuse amalgam we can simply call the national mood" (Ladd, 1992, p. 3).[2] In short, public sentiment is an indicator or summary of how citizens feel about the state of the country, including the government, the economy, and a broad range of other national issues—from crime to pandemics.

In addition to these material factors, news media almost certainly play an important role in shaping how citizens feel about the state of the country. Most people have no direct experience with or information about current events on the national level; their societal perceptions and sociotropic attitudes are likely heavily influenced by media content (Boukes, Damstra, & Vliegenthart, 2021; de Vreese & Boomgaarden, 2003; Shehata, 2021). Even the overall tone of media—the extent to which news is generally positive or

[1] Note that we use the terms "sentiment" and "tone" rather than "mood"—in part to distinguish our focus from well-established measures of policy mood in studies of public opinion (see, e.g., Stimson, 1991, 2024).

[2] While our focus here is on public opinion polls, a great deal of recent work aims to understand public sentiment by extracting text data from social media (see, e.g., Rodríguez-Ibánez, Casánez-Ventura, Castejón-Mateos, & Cuenza-Jiménez, 2023). Early work in this area found close connections between social media sentiment and economic conditions (Daas & Puts, 2014; Lansdall-Welfare, Lampos, & Cristianini, 2012), presenting some indirect evidence that social media sentiment could be similar to the quantity identified in public opinion polls. However, more recent work that formally compares sentiment in survey measures and social media finds very limited connections between the two (Conrad et al., 2021). Although that work is focused on the economy, it nonetheless suggests that sentiment captured in public opinion polls is likely distinct from social media sentiment.

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negative—is correlated with public perceptions of and attitudes about policy issues, the economy, candidates, and governments (e.g., Soroka, 2014). To what degree, then, is the sentiment of the country attributable to the overall tone of public affairs news coverage?

This question assumes that media coverage causes changes in public sentiment, of course. Based on the media effects paradigm, most work in political communication shares this assumption, arguing that information flows unidirectionally from news media to the public. Yet, it seems very likely that media–opinion relationships are in many cases reciprocal. Citizens do learn about politics from the media, but journalists write not just *for* but also *about* their audience (Chadwick, 2011; Krupnikov & Ryan, 2022; Lavrakas, Traugott, & Miller, 2019; McGregor, 2019; Mutz, 1992; Neuner, Soroka, & Wlezien, 2019). This is evident in coverage of the economy, for instance, which not only influences but is also influenced by public perceptions (Soroka, Stecula, & Wlezien, 2015). In fact, recent work suggests that, in some domains, media may be more attentive to public opinion than public opinion is to media (Wlezien, 2024; Wlezien & Soroka, 2024). Thus, it is possible—perhaps even likely—that public sentiment about the state of the country and public affairs news coverage are reciprocally influential, both affecting and reflecting each other.

Our objective is to systematically explore the correlation between public and media sentiment alongside the possibility that the relationship between the two varies across time. We regard the latter as especially important. Questions about public sentiment have been asked for decades, and because media content is also available for decades, we are able to observe public–media relationships over an extended period of time. This offers the opportunity to consider what we regard as an underappreciated aspect of media–public relationships, namely, the likelihood that relationships between media content and public attitudes vary over time. To be clear, there are many papers seeking to determine whether media follows the public or the public follows media, and the degree to which one or the other is true. Findings vary (cf., e.g., Boukes et al., 2021; de Vreese & Boomgaarden, 2003; Wlezien, 2024). Our contention is that this is due in part to the fact that there are times when the connection between media content and the public is strong and times when it is weak; there may also be times that media lead the public and times that media follow.

This paper is thus in large part a methodological exercise, seeking to better understand public mood, media sentiment, and the time-varying relationship between these two measures. We proceed in three steps, focusing on the United States due to its rich history of polling and the long-term availability of U.S. media data.

First, we develop a public opinion measure based on thousands of polls collected between 2000 and 2021 that ask a general "state of the country" question. Using methods from research on preelection preferences (e.g., Erikson & Wlezien, 1999), we estimate a pooled estimate of "public sentiment" in the United States that adjusts for differences across polling organizations. We consider the degree to which questions about sentiment asked by different polling organizations capture the same underlying quantity. We also examine the ways in which this quantity varies over time.

Second, we apply automated content-analytic techniques to hundreds of thousands of network television news transcripts to capture monthly estimates of "media sentiment." Network news remains among the most important sources of news for people in the United States (Newman, Fletcher, Eddy, Robinson, & Nielsen, 2023), and this data set represents the entire population of newscasts from the major TV networks during the 2000–2021 period. As with public sentiment, we consider differences across outlets and variation over time. We share the raw data, sources, and aggregated trend for the public opinion measure, alongside the sources, dictionaries, and resulting data for the media measure for use in other research, available at https://doi.org/10.7910/DVN/CIHAS6.

Third, we use time series models to examine the correspondence between the two measures to better understand exactly what they capture and how they relate. We find that different polling organizations generate broadly similar estimates of public sentiment and that different TV networks do the same for media sentiment. Both sentiment measures vary substantially over time, but there is only a modest and inconsistent connection between them. When there is a connection, it appears that public opinion responds to prior media coverage, as media coverage is largely unaffected by prior public opinion. This unidirectional relationship stands in contrast to some recent research highlighting reciprocal effects regarding public policy preferences (e.g., Wlezien & Soroka, 2024).

Our results based on sentiment are thus a useful reminder that the nature of media effects varies across topics and levels of aggregation. We suspect this conclusion is uncontroversial. However, our results also highlight a less recognized aspect of media effects, namely, that connections between public opinion and media coverage vary over time. Estimations exploring time-varying parameters reveal periods during which media and public sentiment are clearly connected and periods during which they clearly are not. We regard this consideration of time-varying relationships as an important contribution with broad implications for the study of media effects on—and media responses to—public opinion.

Measuring Public Sentiment

How does the U.S. public feel about the state of the country? Polling organizations have been interested in this question for at least the last five decades. In this section, we aggregate all their work, creating a single continuous measure characterizing what the U.S. public generally thinks about the country at any given time. The resulting series is our measure of public sentiment.

The first step to creating a measure of public sentiment is collecting existing polling data. For this purpose, we assembled over 6,000 individual poll results from the period between 1979 and 2021, relying on four data sources: Real Clear Politics, Ballotpedia, Gallup, and a research assistant tasked with collecting polling data from polls' individual websites. In total, these four sources included polls reported by 105 different organizations. Although question wording varied across these organizations (and over time), most polls asked for participants' evaluations of the state of the country, particularly, their satisfaction with the way things are going or whether the country is heading in the right direction or is "on the wrong track." Details on question wording are included in Supplementary Appendix A and Supplementary Tables A1 and A2.[3] The data, the final series, and the scripts

[3] Publicly available polling data on public sentiment typically only report the sentiment of the public as a whole and not the sentiment of any subgroups. As such, what we present here is a general measure of public sentiment that corresponds to our similarly general measure of media sentiment.

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required to turn the former into the latter are all publicly available at https://doi.org/10.7910/DVN/CIHAS6.

Supplementary Appendix B includes a more detailed account of the decisions made in cleaning and preparing the polling data for analysis. In short, after accounting for collaborations between polling agencies and duplicates across data sources, dropping 169 polls with incomplete information about fielding dates, and dropping polls fielded outside the January 2000 to July 2021 window (before which polling data are extremely sparse and after which we have no media data), our working data set included 3,326 polls from 54 polling houses, with each poll dated to the middle day of its fielding period.

All polls reported between January 2000 and July 2021 are plotted in Figure 1. Rather than attempting to associate each poll in the figure with one of the 54 polling houses, we use shapes and colors to indicate polls only from the four houses for which we have the most data (labeled in the figure), and all other polls are shown as solid light gray circles. Each dot represents "net sentiment" for a single poll, specifically, the percentage of positive responses minus the percentage of negative responses. To characterize the overall trends in the data, we overlay the "Raw Trend" with a dashed gray line derived from a smoothed loess trend with a bandwidth of .05.

Figure 1 shows considerable variation in public sentiment over the two decades. Perhaps most importantly, there was a steep decline from net sentiment of approximately 30 in 2001 and 2002, when 30% more people responded positively than negatively, to about −50 net sentiment in 2007 and 2008. Some of this decline may be attributable to 9/11 and the wars in Iraq and Afghanistan, and then eventually to the 2007–2008 financial crisis. Between 2008 and 2020, there was a slight upward trend in public sentiment but not back to the level of optimism of the early 2000s. And there was, of course, a precipitous drop in sentiment with the onset of the COVID-19 pandemic, although it appears to have recovered in late 2020 and 2021. Overall, the public sentiment series seems to be characterized by two long trends—first down, then up—with more ebb and flow around the trend emerging in the second half of the series. However, it is impossible to make strong claims about the causes of either trend given the highly aggregated nature of the data.

Poll results also vary substantially within each time period. This variation is likely due to three factors: sampling error, design effects, and house effects (Biemer, 2010; Groves, 1989; Groves & Lyberg, 2010; Smith, 1978; Wlezien & Erikson, 2001). Sampling error and design effects are especially difficult to estimate and adjust for because of the lack of detailed information about sampling and weighting designs (see Supplementary Appendix B for further details). Nonetheless, we can adjust for house effects—systematic error introduced by polling organizations' different practices. We estimate the influence of survey houses on our estimates of public sentiment by regressing the poll results on dummy variables for (a) each survey house and (b) each date with at least one poll result (polls are dated to the middle day of fielding; see Supplementary Appendix B). We estimate this regression without an intercept so that the coefficients for each date represent daily house-adjusted estimate of public sentiment.

An ANOVA based on this regression model confirms our graphical intuition: estimates of public sentiment vary substantially both over time and across survey houses.[4] How much do estimates vary across houses in practice? Table 1 shows the estimated magnitude of house effects (relative to the median house) of the 10 houses for which we have the most polls. These effects are not an indicator of accuracy or data quality *per se*, they simply reflect the extent to which each house tended to overestimate or underestimate public sentiment relative to other houses. That said, the effects in Table 1 are sizable—Gallup data produce an estimate of sentiment that is roughly 20 points lower than that for CBS or Ipsos, for instance, and more than 30 points lower than SSRS. Despite these substantial differences, Figure 1 suggests that survey firms' results move roughly in parallel over time, intuition that is supported by the ANOVA results and summarized in Footnote 4.

Lacking an indicator of ground truth, in this case, we rely on the wisdom of "the crowd" and use the median house as the basis for adjustment. To account for the variation between houses, we use the model described above to create a measure of sentiment that removes the estimated house effects. This is the "House-Adjusted Trend" shown as a loess-smoothed solid black line in Figure 1. Although the magnitude of some of the house effects in Table 1 are large, the raw and house-adjusted trends are only slightly different, especially after 2005, when multiple organizations regularly report poll results using these questions. Even so, the house-adjusted measure should provide a picture of public sentiment less distorted by individual polling organizations' methodologies and sampling. We accordingly focus on this measure in the analysis that follows though we aggregate the measure by month to match our media data (see Supplementary Appendix B for details on the aggregation process).

Measuring Media Sentiment

Conceptually, our measure of media sentiment should capture the same basic quantity as the measure of public sentiment: the general mood of U.S. news media. More specifically, it should indicate the extent to which the news conveys that things are (or are not) going well. We suspect that the appropriate measure is the sentiment of all public affairs news content. This may be too simplistic—it may be that the media equivalent of public sentiment is some subset, or differentially weighted subsets, of public affairs content. For instance, public sentiment (and the media equivalent) may reflect some weighted combination of news about the economy, foreign affairs, and domestic politics. As a starting point, however, we focus on all public affairs content—that is, all content appearing on major broadcasters' flagship news programs.

We capture content on network television, using all transcripts from ABC, CBS, and NBC's 30-min evening news programs, from 1990 to 2021. Transcripts are available at the story level, with roughly five to seven stories per half-hour program on average, and we aggregate them to the program-day level for analysis. The entire population of transcripts was extracted from Lexis-Nexis, using the now-defunct Lexis-Nexis API, followed by an arrangement directly with the company to provide transcripts in .xml format. The final media sentiment series and the syntax to generate that series—which can be applied to data from Lexis-Nexis or any other large body of media coverage—is publicly available at https://doi.org/10.7910/DVN/CIHAS6.

[4] Time accounts for 90.3% ($p < .001$) of the variation in survey results; house effects account for just 6% of the variation ($p < .001$).

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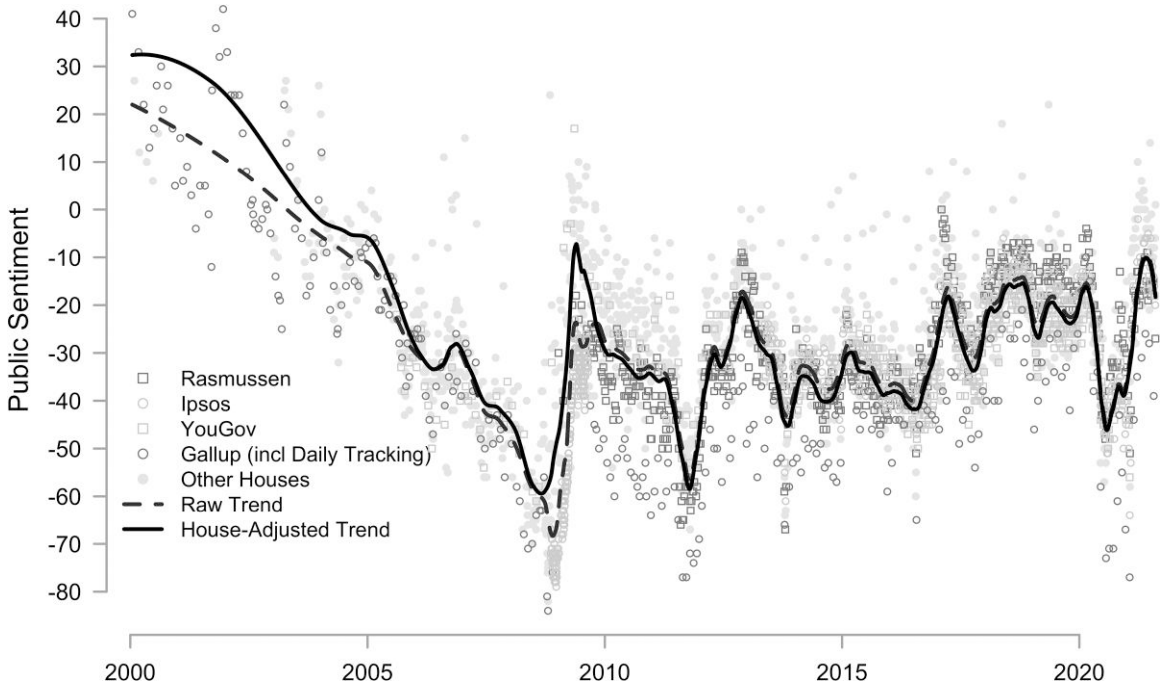


Figure 1. Estimated public sentiment.

Table 1. Estimated House Effects on the Measurement of Public Sentiment

House	Effect
Gallup Daily Tracking	−22.520
Gallup	−17.499
CBS/NYT	0.000
Ipsos	0.302
Morning Consult	0.396
Hart (D), POS (R)	1.556
Rasmussen	2.755
YouGov	3.143
Winning the Issues	9.626
SSRS	12.305

We exclude the major cable networks—CNN, Fox News, and MSNBC—and focus on the "Big Three" television networks for three reasons. First, network news remains the most watched and trusted television news sources for people in the United States, beating local news, cable, and only rivaled in reach by online news sites as a whole (Newman et al., 2023). That is not to say that *everyone* watches network news, but it remains the primary source of news for the modal American. Second, we also expect our measure to provide a good representation of the broader media signal. For instance, research shows that television news coverage closely parallels newspaper coverage, at least in certain policy domains (Soroka & Wlezien, 2022). Although we focus entirely on the Big Three, then, there is good reason to believe that this measure captures trends in the sentiment of other news outlets as well. Third, ABC, CBS, and NBC follow a similar, structured format that is unlike cable news broadcasts. Cable television broadcasts are not typically separated into distinct stories; they also tend to be more partisan and more opinion- and emotion-laden (see, e.g., Atkinson, 2017; Berry & Sobieraj, 2014; Mutz, 2015; York, 2013). Cable news is unquestionably an interesting and important component of the U.S. media system. Limiting our analysis to network news, however, reduces our operational ambiguity and makes clearer the nature of media sentiment as we capture it.

We aggregate the network news data by month to account for noise at the daily level and then capture the sentiment of each month's coverage using the Lexicoder Sentiment Dictionary (LSD), a simple dictionary-based approach to sentiment analysis initially designed for and tested on North American news content (Young & Soroka, 2012). The LSD has been described in detail and tested against both human coding and other dictionaries using U.S. news content (Young & Soroka, 2012). It has been further tested against other dictionaries in subsequent publications as well (e.g., Soroka et al., 2015; Wlezien, Soroka, & Stecula, 2017). We accordingly rely on this prior work as confirmation of the reliability and validity of the dictionary.

The implementation of the LSD is simple: it captures the number of positive words and the number of negative words in a news article.[5] "Net" sentiment is then estimated using a simple measure proposed by Lowe, Benoit, Mikhaylov, and Laver (2011) and used in Proksch, Lowe, Wäckerle, and Soroka (2019): log [(pos. counts + 0.5)/(neg. counts + 0.5)]. This is an empirical logit, slightly smoothed toward zero, and very highly correlated with another standard approach: (pos. counts – neg. counts)/total words. The resulting net sentiment is greater than zero when positive words outnumber negative words and less than zero when negative words

[5] We rely on the implementation of the LSD in the quanteda package in R (Benoit et al., 2018). That dictionary includes 2,858 negative words such as "breakdown*", "contaminat*", "downtrodden", "jeer*" and "rotting"; and 1,709 positive words such as "admiring*", "brave*", "honesty", "lucrative*" and "splendor". Each of these words is tested in Young and Soroka (2012), as is the tendency for the balance of positive and negative words to be correlated with human-coded sentiment of news articles.

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outnumber positive ones. See Footnote 6 for additional detail on interpreting the scale.

There are two issues worth nothing where the LSD is concerned. The first is whether any measure of generalized sentiment is useful in this research context. Given their availability, sentiment lexicons are now widely used, sometimes in analyses that would benefit from a measurement strategy more directly capturing the quantity of interest (see, e.g., Bestvater & Monroe, 2023). In this instance, however, general sentiment *is* our quantity of interest—we care about the broad optimism or pessimism in media coverage, and we regard the general sentiment of this coverage as the most relevant quantity for comparison to our measure of public mood.

The second is whether a dictionary-based approach to capturing sentiment is sufficiently accurate, relative to other approaches to sentiment analysis, including models designed to "learn" from large bodies of human-coded data which then apply those codes to far larger data sets. Some recent works find important gains from machine-learning vis-à-vis dictionary-based analyses, although these gains are context-dependent and can be relatively limited once article-level results are aggregated (see, e.g., Alantari, Currim, Deng, & Singh, 2022; Dun, Soroka, & Wlezien, 2021; Reveilhac & Morselli, 2022; van Atteveldt, van der Velden, & Boukes, 2021). Especially for the kind of generalized sentiment we are interested in here—captured across many new stories over an extended period of time—we regard a dictionary approach as appropriate; that said, we revisit possible alternative measures of sentiment in the concluding section.

Figure 2 shows estimates of media sentiment using television transcripts from the three major networks and employing the method outlined above. Circles show monthly estimates for each network separately; the black line shows a loess-smoothed trend with a .05 bandwidth. As with public sentiment in Figure 1, we see that media sentiment sharply declines in the early 2000s from about −.2 to −.4, perhaps due to 9/11 and the subsequent wars in Iraq and Afghanistan.[6] Media sentiment gradually improves to approximately 0 between 2003 and 2009, a time during which public sentiment becomes more negative; indeed, prior work suggests that media coverage tends to return to a less negative equilibrium during long periods of "real-world" negativity (Soroka & Krupnikov, 2021). During the Great Recession and the Obama administration, media sentiment was slightly negative, but sentiment declined again to about −.3 during the last 2 years of Obama's presidency and remained more negative under Trump. Perhaps counterintuitively, and quite unlike public sentiment, media sentiment dramatically *improved* shortly after the onset of the COVID-19 pandemic. Here, it returns to 0, the highest value we see across the whole series, similar to just before the Great Recession. Again, this is in line with prior work on media equilibriums during prolonged negative conditions. Overall, while public sentiment is characterized by long trends, media sentiment seems to involve shorter-term, possibly cyclical variance around a slower-moving baseline.

To what degree are there differences in the sentiment between networks? The average interitem correlation between the three network measures 0.48. In an ANOVA of monthly sentiment modeled as a function of time and network, time accounts for 85.3% of the variation in media sentiment, while network accounts for 12.6%—roughly parallel to the house effects ANOVA conducted for public sentiment. There clearly is a general "media sentiment" that shifts over time in roughly parallel ways across TV networks in the United States, but there is some independence in network coverage as well. That is not to say that the version of media sentiment reflected in network news is necessarily similar to cable news programming or online news, although there is some evidence of parallelism in policy coverage across platforms (Soroka & Wlezien, 2022). Regardless, there is clearly overtime variation in the tenor of evening network news. Next, we consider whether that variation in media sentiment links to variation in public sentiment.

Are Public Sentiment And Media Sentiment Related?

Having generated measures of both public and media sentiment, we now examine the degree to which the two measures are related. Even a cursory consideration of the series shown in Figures 1 and 2 suggests a rather weak correlation between net sentiment in the public and media coverage. In fact, the raw correlation between the monthly series is negative ($r = -.362$, $p < .001$). This is driven largely by the steady downward trend in public sentiment from 2000 to 2009, even as media coverage trends upward in the latter half of that time period. What accounts for the steady decay in public sentiment, leading to a new equilibrium in the series from 2010 onward? There appears to be some decline even before 9/11 (although we have only 18 months of data in that window), so it seems unlikely that 9/11 is a complete explanation for the decay in sentiment. That said, it may be that 9/11 intensified an already ongoing decline in sentiment, attributable to a number of different political events at the time, including controversies about ballots in the 2000 election and other events over the course of the ensuing Bush presidency. Our data cannot provide much of an explanation on this front, however, and the raw correlation between public and media sentiment remains weak when we restrict the analysis to the period from 2009 onward ($r = .008$, $p = .924$). Even after the secular decline in public sentiment, then, a connection between the series of public and media sentiment is not evident.

That said, it may be that we should not expect a relationship between the *levels* of public preferences and media coverage. Previous research shows that economy-related news tends to focus on change (Soroka et al., 2015; Soroka & Wlezien, 2022; Wlezien et al., 2017). To the extent in which media reflect public opinion, we might accordingly expect media sentiment to be generally positive when public opinion is becoming more optimistic and generally negative when public opinion is becoming more pessimistic, regardless of the actual level of public sentiment. That is, we might instead expect an association between levels of coverage and *changes* in public opinion. There are other possibilities as well, of course; and rather than settling this by assumption, we can do so empirically by using a general modeling approach (De Boef & Keele,

[6] Note that our logged measure of net sentiment (described above) weighs individual words more heavily when overall counts are low. For instance, 0 positive words and 1 negative word produce a net sentiment score of −.48, while 2 positive words and 3 negative words produce a net sentiment score of −.15. This is by design, since it takes into the account the impact of a valenced word relative to the total number of valenced words. This makes it difficult to directly interpret a drop from −.2 to −.4. As a rough approximation, this drop is equivalent to a shift from a 1 positive word and 2 negative words (−.22) to 1 positive word and 3 to 4 negative words (−.37 and −.48, respectively).

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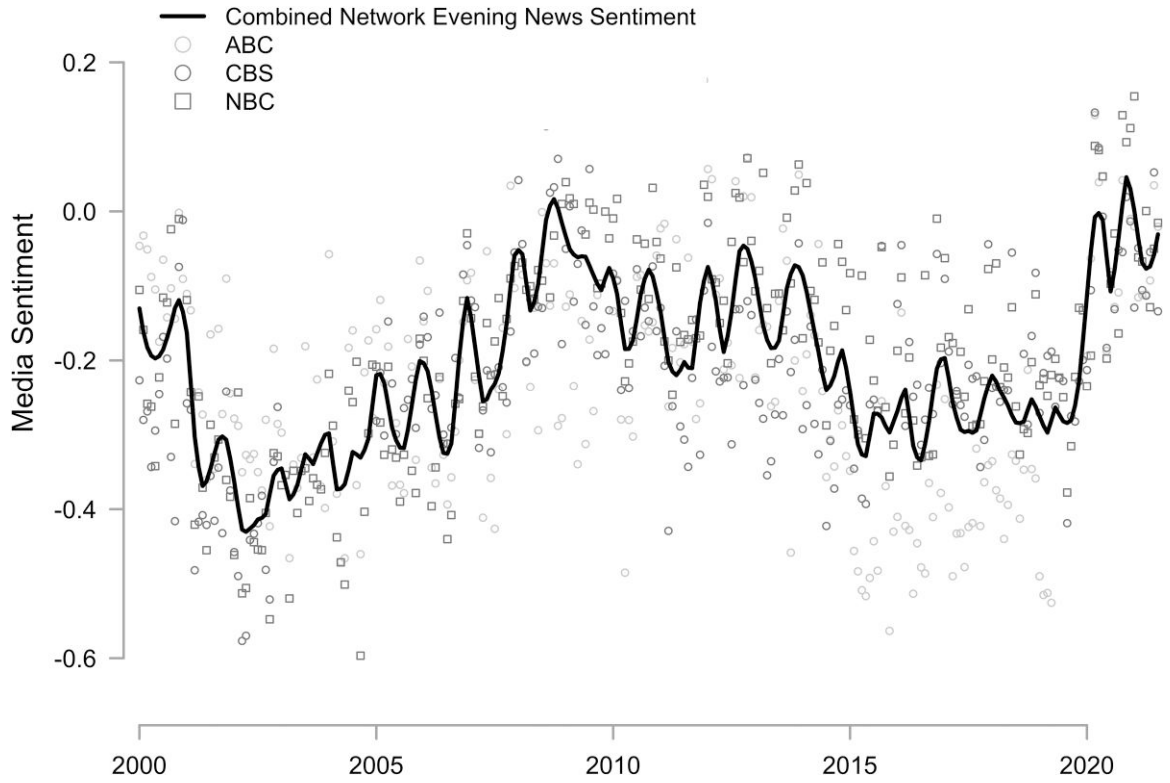


Figure 2. Estimated media sentiment.

2008). In that sense, our analysis is fundamentally exploratory. We aim to describe the trends present in the data and uncover any evidence of contemporaneous or lagged relationships and hope that this is informative, serving as a basis for future research.

For this, we employ error correction models (ECMs), in which changes in one variable are regressed on lagged levels of that variable and both current changes and lagged levels of the other variable. We opt for ECMs because they allow us to identify a range of different relationships that might be present in the data that other approaches settle by assumption (De Boef & Keele, 2008).[7] Importantly, ECMs clearly isolate different quantities of special interest, including the effects of current changes (i.e., time-serial "differences") in variables versus effects associated with lagged levels. The specific equations are as follows:

$$\Delta\,\text{Media}_t = b_0 + b_1\text{Media}_{t-1} + b_2\Delta\text{Public}_t + b_3\text{Public}_{t-1} + {}^{\text{M}}\text{e}_\text{t} \quad (1)$$

$$\Delta\,\text{Public}_t = b_4 + b_5\text{Public}_{t-1} + b_6\Delta\,\text{Media}_t + b_7\text{Media}_{t-1} + {}^{\text{P}}\text{e}_\text{t} \quad (2)$$

where the delta (Δ) symbol designates changes and our temporal (t) units are months. These equations capture the responsiveness of each of our variables to both current changes and lagged levels of the other variable. We emphasize that coefficients for current changes are essentially contemporaneous relationships and do not provide much evidence about the directional relationship between the variables. In Equation 1, for instance, b_2 may represent the effect of public opinion on media coverage or the effect of media coverage on public opinion, as both are measured at the same point in time. The effect (b_3) of lagged levels of opinion provides somewhat stronger evidence of a potential directional relationship, as there is no real basis for suspecting that future changes in media coverage would cause previous levels of public opinion.[8,9] However, the estimated effect of lagged public opinion is a conservative one. Directional relationships between these series may exist on shorter time scales than the monthly lags on which we rely here, in which case the real impact of public opinion on later news coverage may be understated because it is at least partly reflected in previous media coverage. The same logic applies for estimated effects of lagged media on public opinion. As such, the likelihood of Type II errors for lagged coefficients is almost certainly inflated to some degree, so we may well observe null effects for the lagged coefficients even when the true effects of the variable are nonzero. Given this, any evidence of these lagged relationships should be taken quite seriously.

To estimate Equations 1 and 2 and interpret the results, it is important that both sides of the equations are stationary—that the two variables tend to have constant means over the same time period. However, Figure 1 shows that public sentiment is clearly nonstationary: the series declines for 8 years,

[7] Autoregressive distributed lag models are a mathematically equivalent approach, and we opt for ECMs because they more clearly represent different "effects" of interest.

[8] In principle, public opinion surveys fielded over long periods of time might spill across months, biasing b_3 downward and biasing b_1 upward (i.e., increasing time-serial dependency). However, there are only 12 polls fielded for 15 days or longer, so this possibility is unlikely to meaningfully bias our coefficients.

[9] The specification of cross-lagged effects in Equations 1 and 2 thus encompasses Granger causality.

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Table 2. Error Correction Models of Media and Public Sentiment

	2000–2021, DV:		2009–2021, DV:	
	Δ Media$_t$	Δ Public$_t$	Δ Media$_t$	Δ Public$_t$
Δ Media$_t$	—	0.032	—	0.115^{+}
		(0.054)		(0.060)
Media$_{t-1}$	−0.299***	0.001	−0.342***	0.030
	(0.045)	(0.042)	(0.062)	(0.050)
Δ Public$_t$	0.044	—	0.209^{+}	—
	(0.073)		(0.110)	
Public$_{t-1}$	−0.012	−0.205***	−0.001	−0.126***
	(0.047)	(0.038)	(0.057)	(0.041)
Constant	−0.484***	0.007	−0.467***	0.046
	(0.086)	(0.078)	(0.101)	(0.080)
N	258	258	151	151
Rsq	0.150	0.102	0.193	0.085

Note. Cells contain OLS regression coefficients with standard errors in parenthesis.
$^{+}p < .10$.
$^{*}p < .05$.
$^{**}p < .01$.
$^{***}p < .001$.

after which it tends to be stationary.[10] In contrast, although media sentiment drifts up and down for extended periods, diagnostic tests suggest the overall series is stationary.[11]

Results from estimating both equations over the entire time period are displayed in the first two columns of Table 2. To account for different metrics and variances in the two sentiment measures, we standardize both rescaling them to indicate standard deviations from 0. We account for nonstationarity in the first 8 years of the public sentiment series by detrending the series prior to 2008.[12] In the first column, the coefficient for changes in public sentiment is positive but not significant; the same is true for the coefficient for changes in media sentiment in the second column. Thus, the results suggest no relationship between current changes in media sentiment and current changes in public sentiment. There also is no evidence of an impact of lagged public sentiment on media (column 1) nor of lagged media sentiment on the public (column 2).

The steep downward trend in public sentiment over the first 8 years in Figure 1 is partly "corrected" when we use a detrended measure, but it may nevertheless be the case that the relationship between media and public sentiment is different once both are moving around a stable equilibrium. As a preliminary test of this possibility, the third and fourth columns of Table 2 reestimate the equations focusing just on data from 2009 onward. There are some important differences between the estimations. Focusing on the latter time period leads to weak ($p < .10$) evidence of connections between the two series: changes in public sentiment positively predict changes in media sentiment (column 3), and changes in media sentiment positively predict changes in public sentiment (column 4).

There is some indication of a reciprocal relationship between measures of public and media sentiment in Table 2—but weak and only in the second decade. The possibility that the relationship between the series may vary more generally over time is our next focus. We explore this potential variation by estimating models over moving "windows" of the full time series. To the extent that relationships vary over time, so too should the estimated coefficients. For our analysis, we focus on 60-month intervals, beginning with a window that starts in January 2000 and ends in December 2004. We then move the window forward 1 month, by dropping January 2000 and adding January 2005. We repeat this for the entire time series, using the same measures (standardized and detrended where necessary) as in Table 2.

We plot the resulting coefficients in Figures 3 and 4. Figure 3 focuses on Equation 1, capturing whether current changes and lagged levels of public sentiment predict current changes in media sentiment. The top panel of Figure 3 plots time-varying coefficients and 95% confidence intervals for the impact of current *changes* in public sentiment. The bottom panel plots the corresponding estimates and confidence intervals for the impact of lagged *levels* of public sentiment. The coefficients are shown at the start of each interval, so that the first coefficient is the relationship between public sentiment and media sentiment for the 60 months starting in January 2000.

The top panel of Figure 3 suggests that changes in public sentiment were non-significantly negatively associated with changes in media sentiment at the beginning of the time period. A positive association between public sentiment and media sentiment emerges in 2003, and this becomes statistically significant in 2007. Thereafter, in some but not all periods, there is a significant positive relationship between changes in public sentiment and changes in media sentiment. By contrast, the bottom panel of Figure 3 shows that the coefficients for lagged levels of public sentiment mostly hover around 0 and are essentially never significant. Thus, while there is some evidence of a positive association between changes in public

[10] Although the Dickey–Fuller (DF) statistic (−2.938) implies that we may be able to reject the null of a unit root ($p = .04$), the coefficient (−0.05) for lagged opinion is substantively close to 0 and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test rejects stationarity ($p < .01$). After the long decline in sentiment is complete in 2008, however, the series appears to be stationary (augmented DF test $p = < .01$).

[11] Augmented DF tests give marginal evidence against a unit root ($p = .07$), but the lagged media coefficient (−0.14) is substantively different from 0 and KPSS tests are confirming ($p = .10$).

[12] We do so by regressing public sentiment prior to 2008 on a time counter variable and using the residuals (the variation not accounted for by a linear over-time trend) in place of the original series.

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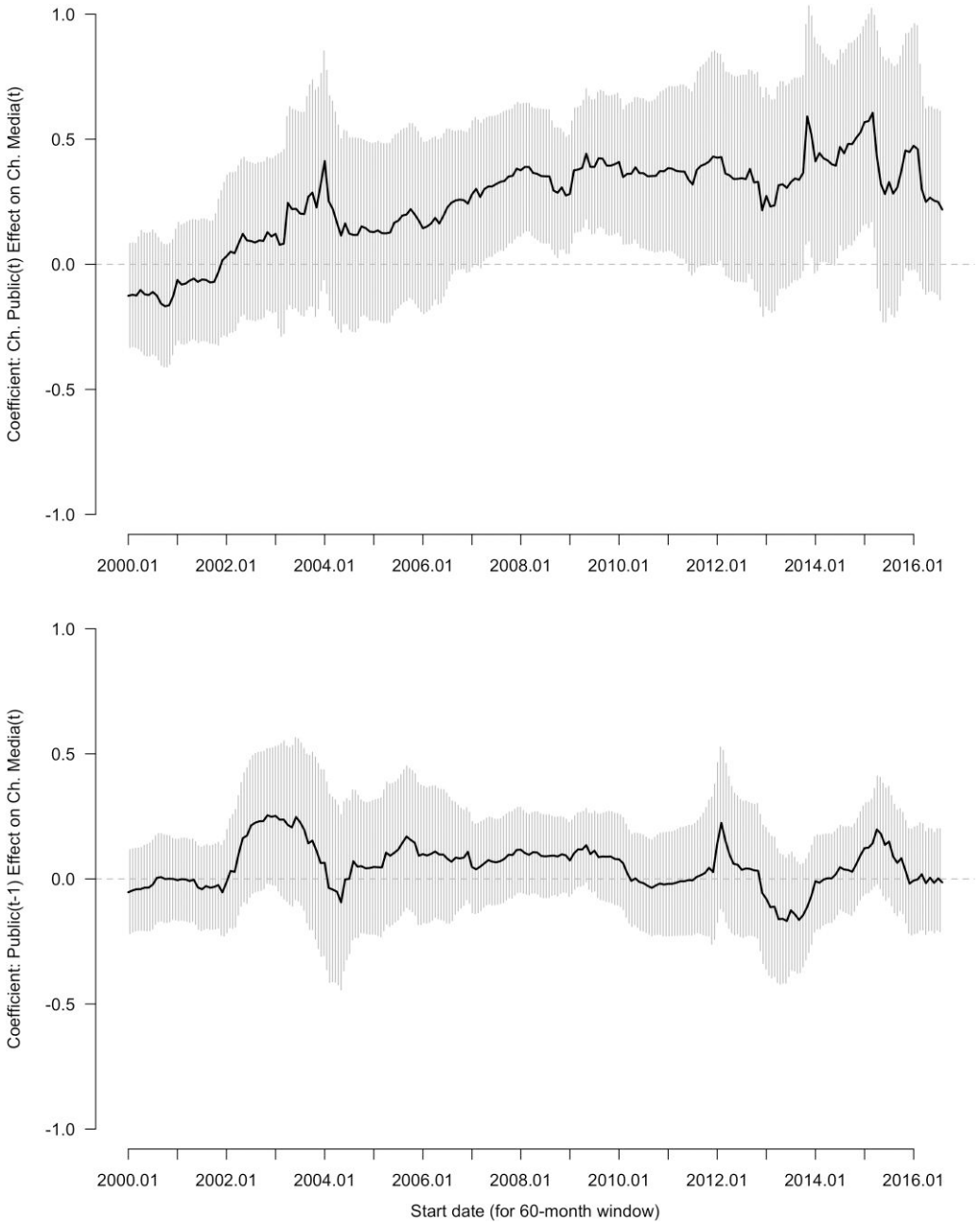


Figure 3. *The estimated impact of public sentiment on media sentiment.* Black lines represent coefficients for each 60-month interval and shaded areas represent 95% confidence intervals.

sentiment and changes in media sentiment, the relationship appears entirely concurrent. There is little basis for inferring a causal effect, recalling that any contemporaneous relationships may be bidirectional. Based on the results, we find no evidence that the general sentiment of media coverage responds to public opinion.

What about media effects on public sentiment? Figure 4 plots whether current changes and lagged levels of media sentiment predict changes in public sentiment, following Equation 2, again using 60-month moving windows. In the top panel of the figure, we can see that the estimated relationship between changes in the tone of current media coverage and changes in public sentiment parallels what we saw in Figure 3. That is, the coefficients are slightly negative early in the timeline but then increase, becoming positive after 2003 and significantly different from 0 after 2006. Thereafter, we see mostly significant positive estimates. Taken together with what we see in Figure 3, the results underscore the existence of a relationship between contemporaneous shifts in public and media sentiment in each month, the strength of which varies over time. However, as with the top panel of Figure 3, these results provide little basis for inferring (or even seriously contemplating) causal effects in one direction or the other.

The estimated relationships between public sentiment and lagged media sentiment in the bottom panel of Figure 4 are more compelling. Here, we see evidence suggestive of a causal relationship, as levels of media sentiment predict changes in public sentiment in the subsequent month—albeit only significantly so when the start date for the 60-month window is

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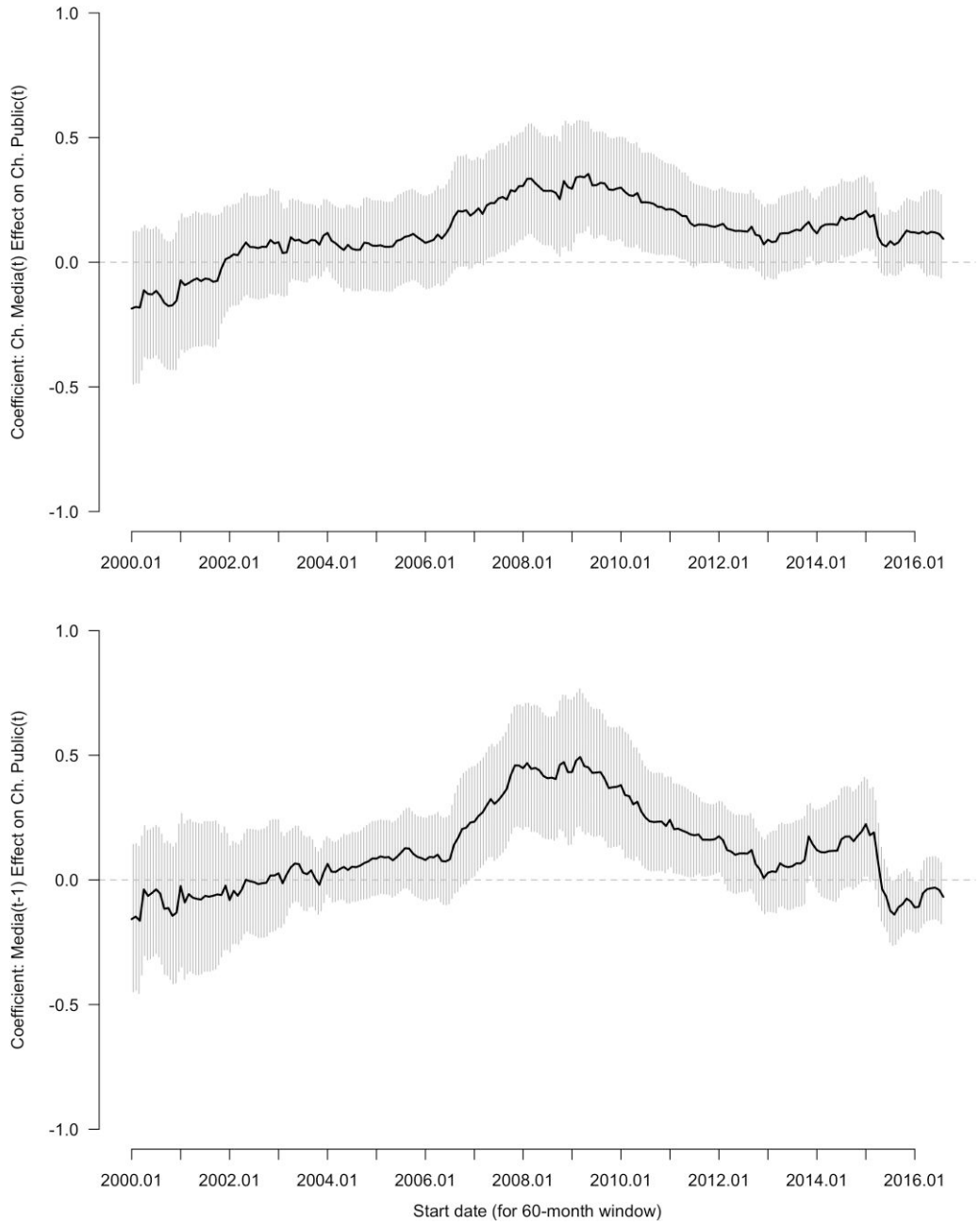


Figure 4. *The estimated impact of media sentiment on public sentiment.* Black lines represent coefficients for each 60-month interval and shaded areas represent 95% confidence intervals.

between 2007 and 2012. This stands in contrast to the results in the bottom panel of Figure 3, which provide no evidence that levels of public sentiment predict subsequent media changes. Results in Figure 4 thus imply that media content may influence public sentiment, at least later in the timeline. In other words, influence appears to run mostly from media to the public, not the other way around. Of course, the concurrent relationship shown in the top panels of Figures 3 and 4 suggest that there may be contemporaneous reciprocal causation as well, and public opinion may have effects on media coverage at a shorter time scale than we can evaluate here. Nonetheless, our analyses suggest that media is more likely to lead than follow public sentiment, although this varies over time.

Discussion

We have developed and examined novel measures of public and media sentiment between 2000 and 2021. We focus on sentiment in the United States, an ideal case for our investigation due to the long-term availability of both polling and media data. Although there is some variation in both measures that can be attributed to differences across polling houses and television networks, we show that both measures vary much more over time than across organizations. For scholars of public opinion and/or media content, these findings are important: different survey questions from different polling organizations appear to capture roughly similar overtime variation in public sentiment, and trends in the

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sentiment of media coverage are similar across different media outlets.

For public sentiment, it is thus not only possible but also fruitful to create aggregated, national-level measures that track public evaluations of the state of the country over the last two decades. We believe the resulting measure is a useful contribution for scholars interested in broad indicators of public attitudes, and we hope that other scholars extend our approach to examine sentiment in other countries. Yet, Figure 1 suggests that this measure may raise as many questions as it answers. The steep downward trend in public sentiment between 2000 and 2008 may not be surprising given the occurrence of 9/11 and subsequent wars. As we noted earlier in the paper, however, the downturn *precedes* 9/11. Then, between 2008 and 2010, in the depths of the Great Recession, the series regains nearly half of the positivity lost over the preceding decade. Something about the underlying logic or contents of public sentiment appears to have changed in 2008 or 2009, co-occurring with Obama's election but persisting into the Trump and Biden presidencies as well. We have accounted for this in our analyses by detrending the first half of the time series, but we cannot yet offer a substantive explanation for what appears to be a fundamental shift in the nature of general public sentiment in the 2000s.

For media sentiment, we have developed a measure tracking the overall tone of U.S. evening network newscasts since 2000. Unlike the public opinion measure, the media measure behaves much as we would expect—we observe changes in tone during pivotal moments in U.S. political history, including 9/11, presidential elections, and the outbreak of the COVID-19 pandemic. Future work might use this measure for similar investigations of overtime dynamics related to media content, examination of how media sentiment varies over time across different countries with different media systems, or simply as a baseline for comparison and refinement with different content-analytic approaches to sentiment analysis, including machine-learning approaches.

Our primary focus is on how much public sentiment and media sentiment covary. The answer is nuanced. After detrending the public sentiment series, we find a positive association between contemporaneous changes in public sentiment and changes in media sentiment, consistent with the observation that both media and the public tend to focus on changing conditions and novel information (e.g., Soroka et al., 2015; Soroka & Wlezien, 2022; Wlezien et al., 2017). However, this relationship is sizable and statistically significant only when the start date for the 60-month window is between about 2007 and 2012. The varying significance of the relationship over the time period examined here is rarely identified in research focused on public opinion and media coverage or, indeed, in social science more generally (see Munger, 2023). There clearly are periods during which the public and media move together, perhaps simultaneously responding to similar phenomena, and other periods when this is not the case. Prior work has focused relatively little on time-varying connections between the public and media, but our results suggest that this variation may be an underappreciated aspect of public–media relationships.

We have also considered the directionality of the relationship between public and media sentiment. Our analyses provide no evidence that media respond to public sentiment, but they do suggest that the public responds to the sentiment of media coverage. Again, this is true only for the 2007–2017 period, so public opinion does not *always* respond to media sentiment. This conclusion may be unsurprising to those skeptical of strong media effects in a fragmented media environment. By the 2000s, network news in the United States had already lost its monopoly on national-level information to competition from cable news, and its sphere of influence continued to shrink throughout the 2000s and 2010s with the increasing availability of online information. This is not to say that network news became irrelevant—it retains the single largest audience of any traditional news platform. Even so, we might expect the effects of network news to weaken during this time period. Yet, we find that they instead become stronger: it is only after 2007, in a fully digital world, that the sentiment of network news significantly predicts later public sentiment. This is not due to stable estimates of a positive relationship becoming increasingly precise (due to, e.g., increasing quantities of polling data)—the coefficient itself increases up to a substantively large effect size ($d = 0.5$) and declines quickly afterward. The contemporaneous relationship between the series is never more than a comparatively modest $d = 0.25$, but it is clear that media effects on public opinion are neither trivial nor monotonically weakening as the media environment continues to fragment.

In light of this temporal variation, we view our results as providing modest evidence of a unidirectional effect of media on public attitudes,[13] consistent with long-standing intuitions about media effects but in direct contrast to recent work on reciprocal media–public dynamics (e.g., Wlezien, 2024; Wlezien & Soroka, 2024). Of course, our findings are not dispositive about directionality. If media responds more quickly to changes in public sentiment than the public responds to media, our monthly analysis may underestimate media responsiveness that occurs at, say, the daily or weekly level. The nature of sentiment may also account for this contrast: it is more amorphous than evaluations of the economy, specific public policies, or political candidates. Indeed, it is not clear what elements of public opinion that media coverage would attend to and follow. The media signal we have identified is relatively clear and broadly salient because it is the sum of all public affairs coverage. Our measure of public sentiment is more opaque. We cannot identify which considerations survey respondents draw on when they answer these questions, only that different questions asked by different polling organizations identify a similar underlying quantity. Public opinion may vary widely across domains unrelated to those captured by the state of the country questions that compose our measure of public sentiment, and media may only attend to some of these domains.

Indeed, this difference between public and media sentiment might partially explain the time-varying nature of the public–media relationships. If our focus on overall sentiment misses a more nuanced dynamic in which the public cares not so much about the general state of the country but about specific issues or topics, we might expect a strong relationship between public sentiment and media coverage when the two align in focus, with media extensively covering the matters the public cares

[13] Of course, lagged relationships alone are not definite evidence of causality. They could indicate, for instance, that the two series respond at different speeds to the same underlying changes in real-world conditions. However, if different response speeds explained the relationships we see in our data, we would expect media to lead public opinion consistently across the whole time period, rather than just from 2007 to 2017. Thus, there is reason to think that public opinion is actually following media sentiment, at least in this period.

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most about. The 2007–2017 period in which we observe a relationship between the two series partially coincides with the Great Recession, when the U.S. public deemed the economy by far the most important issue facing the country (Gallup, 2024). Moreover, the proportion of news stories about the economy each month (Supplementary Figure C1) suggests that media were also highly attentive to the economy in this period, albeit less enduringly so than the public. This seems like fertile ground, then, for the sort of public–media alignment that might lead to a strong relationship between the two series.

However, when we reestimate our analyses focusing just on the sentiment of economic articles, rather than all public affairs news content as before, we find no evidence of connections between public and media sentiment (see Supplementary Appendix C and Supplementary Figures C2 and C3 for these results). Even during a historic recession when both the public and the media were highly focused on the economy, the broader relationships between the two series demonstrated in Figures 3 and 4 reflect more than shared responses to economic conditions. This rules out the most intuitive explanation, leaving open the question of what contextual factors account for the time-varying public–media relationship. This question stands out as an especially important area for future research aimed at understanding social contexts in which media effects and media responsiveness to public opinion are more or less likely.

In short, there is much more to learn about the nature of public and media sentiment and their associations, both in the United States and in other countries. Exploring many of the issues—about the considerations the public draws on when answering state of the country questions, topical variation in news coverage and domain-specific relationships, differences between media systems, and so on—is certainly feasible given the ready availability of news content and our novel measure of public sentiment. For the time being, however, we regard the sentiment measures themselves and our emphasis on time-varying relationships as broadly useful conceptual and methodological contributions to the study of media and public opinion.

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Supplementary Data

Supplementary data is available at *IJPOR* online.

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Data Availability

The aggregated measures of public and media sentiment and code to generate analyses and figures are available at doi:10.7910/DVN/CIHAS6. We also provide the raw polling data (except for proprietary data from Gallup) and the script used to process polls into the final public opinion measure. We cannot provide the raw media data, which are copyrighted, but we provide information allowing researchers to replicate our search.

References

Alantari, H. J., Currim, I. S., Deng, Y., & Singh, S. (2022). An empirical comparison of machine learning methods for text-based sentiment analysis of online consumer reviews. *International Journal of Research in Marketing*, 39, 1–19. doi:10.1016/j.ijresmar.2021.10.011

Atkinson, M. L. (2017). *Combative politics*. Chicago, IL: The University of Chicago Press.

Basu, Z. (2024 , August 10). Harris vs. Trump. *Axios*. Retrieved from https://www.axios.com/2024/08/10/harris-walz-joy-trump-revenge

Benoit, K., Watanabe, K., Wang, H., Nulty, P., Obeng, A., Müller, S., & Matsuo, A. (2018). quanteda: An R package for the quantitative analysis of textual data. *Journal of Open Source Software*, 3, 774–774. doi:10.21105/joss.00774

Berry, J. M., & Sobieraj, S. (2014). *The outrage industry*. Oxford, NY: Oxford University Press.

Bestvater, S. E., & Monroe, B. L. (2023). Sentiment is not stance: Target-aware opinion classification for political text analysis. *Political Analysis: An Annual Publication of the Methodology Section of the American Political Science Association*, 31, 235–256. doi:10.1017/pan.2022.10

Biemer, P. P. (2010). Total survey error: Design, implementation, and evaluation. *Public Opinion Quarterly*, 74, 817–848. doi:10.1093/poq/nfq058

Boukes, M., Damstra, A., & Vliegenthart, R. (2021). Media effects across time and subject: How news coverage affects two out of four attributes of consumer confidence. *Communication Research*, 48, 454–476. doi:10.1177/0093650219870087

Chadwick, A. (2011). The political information cycle in a hybrid news system: The British prime minister and the "bullygate" affair. *The International Journal of Press/Politics*, 16, 3–29. doi:10.1177/1940161210384730

Conrad, F. G., Gagnon-Bartsch, J. A., Ferg, R. A., Schober, M. F., Pasek, J., & Hou, E. (2021). Social media as an alternative to surveys of

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opinions about the economy. *Social Science Computer Review*, 39, 489–508. doi:10.1177/0894439319875692

Cook C. (2020, July 16) Right direction/wrong track numbers tell the story of the election. *National Journal*. Retrieved from https://www.nationaljournal.com/s/708437/right-directionwrong-track-numbers-tell-the-story-of-the-election

Daas, P. J. H., & Puts, M. J. H. (2014). *Social media sentiment and consumer confidence* (Working Paper 5). ECB Statistics Paper. doi:10.2866/11606

De Boef, S., & Keele, L. (2008). Taking time seriously. *American Journal of Political Science*, 52, 184–200. doi:10.1111/j.1540-5907.2007.00307.x

de Vreese, C., & Boomgaarden, H. (2003). Valenced news frames and public support for the EU. *IEEE Transactions on Communications: IEEE Communications Society*, 28, 361–381. doi:10.1515/comm.2003.024

Dun, L., Soroka, S., & Wlezien, C. (2021). Dictionaries, supervised learning, and media coverage of public policy. *Political Communication*, 38, 140–158. doi:10.1080/10584609.2020.1763529

Erikson, R. S., & Wlezien, C. (1999). Presidential polls as a time series: The case of 1996. *Public Opinion Quarterly*, 63, 163–177. doi:10.1086/297709

Fox News Polling Unit. (2024, November 5). Fox News Voter Analysis. *Fox News*. Retrieved from https://www.foxnews.com/politics/fox-news-voter-analysis-change-election-top-issues-mood-nation-voters-weigh

Gallup. (2024). Most Important Problem. *Gallup.Com*. Retrieved from https://news.gallup.com/poll/1675/Most-Important-Problem.aspx

Groves, R. M. (1989). *Survey errors and survey costs*. New York, NY: Wiley.

Groves, R. M., & Lyberg, L. (2010). Total survey error: Past, present, and future. *Public Opinion Quarterly*, 74, 849–879. doi:10.1093/poq/nfq065

Krupnikov, Y., & Ryan, J. B. (2022). *The other divide*. Cambridge, MA: Cambridge University Press.

Ladd, E. C. (1992). Big swings in the national mood are a staple of contemporary politics. *Public Perspective (January/February)*, 3, 3–5.

Lansdall-Welfare, T., Lampos, V., & Cristianini, N. (2012). Effects of the recession on public mood in the UK. *Proceedings of the 21st International Conference on World Wide Web*, 1221–1226. doi:10.1145/2187980.2188264

Lavrakas, P. J., Traugott, M., & Miller, P. V. (2019). *Presidential polls and the news media*. New York, NY: Routledge.

Lowe, W., Benoit, K., Mikhaylov, S., & Laver, M. (2011). Scaling policy preferences from coded political texts. *Legislative Studies Quarterly*, 36, 123–155. doi:10.1111/j.1939-9162.2010.00006.x

McGregor, S. C. (2019). Social media as public opinion: How journalists use social media to represent public opinion. *Journalism*, 20, 1070–1086. doi:10.1177/1464884919845458

Munger, K. (2023). Temporal validity as meta-science. *Research & Politics*, 10, 20531680231187271. doi:10.1177/20531680231187271

Mutz, D. C. (1992). Impersonal influence: Effects of representations of public opinion on political attitudes. *Political Behavior*, 14, 89–122. doi:10.1007/BF00992237

Mutz, D. C. (2015). *In-your-face politics*. Princeton, NJ: Princeton University Press.

Neuner, F. G., Soroka, S., & Wlezien, C. (2019). Mass media as a source of public responsiveness. *The International Journal of Press/Politics*, 24, 269–292. doi:10.1177/1940161219832416

Newman, N., Fletcher, R., Eddy, K., Robinson, C. T., & Nielsen, R. K. (2023). *Digital news report* 2023. Reuters Institute for the Study of Journalism. doi:10.60625/RISJ-P6ES-HB13

Proksch, S.-O., Lowe, W., Wäckerle, J., & Soroka, S. (2019). Multilingual sentiment analysis: A new approach to measuring conflict in legislative speeches. *Legislative Studies Quarterly*, 44, 97–131. doi:10.1111/lsq.12218

Reveilhac, M., & Morselli, D. (2022). Dictionary-based and machine learning classification approaches: A comparison for tonality and frame detection on twitter data. *Political Research Exchange: An ECPR Journal*, 4, 2029217. doi:10.1080/2474736X.2022.2029217

Right/wrong track polls. (2024, November 27). *The Hill*. Retrieved from https://elections2024.thehill.com/national/right-wrong-track/

Rodríguez-Ibánez, M., Casánez-Ventura, A., Castejón-Mateos, F., & Cuenca-Jiménez, P.-M. (2023). A review on sentiment analysis from social media platforms. *Expert Systems with Applications*, 223, 119862. doi:10.1016/j.eswa.2023.119862

Saad, L. (2025, May 23). U.S. Political, Economic Indicators Steady in May. *Gallup.Com*. Retrieved from https://news.gallup.com/poll/690986/political-economic-indicators-steady-may.aspx

Shehata, A. (2021). Media effects on perceptions of societal problems. In J. Morrison, J. Birks, M. Berry (Eds.), *The Routledge companion to political journalism*. (1st ed. 302–311). London, UK: Routledge.

Smith, T. W. (1978). In search of house effects: A comparison of responses to various questions by different survey organizations. *Public Opinion Quarterly*, 42, 443–463. doi:10.1086/268473

Soroka, S. (2014). *Negativity in democratic politics*. Cambridge, MA: Cambridge University Press.

Soroka, S., & Krupnikov, Y. (2021). *The increasing viability of good news*. Cambridge, MA: Cambridge University Press.

Soroka, S., Stecula, D. A., & Wlezien, C. (2015). It's (change in) the (future) economy, stupid: Economic indicators, the media, and public opinion. *American Journal of Political Science*, 59, 457–474. doi:10.1111/ajps.12145

Soroka, S., & Wlezien, C. (2022). *Information and democracy*. Cambridge, MA: Cambridge University Press.

Stimson, J. A. (1991). *Public opinion in America*. Boulder, CO: Westview Press.

Stimson, J. A. (2024). *The origin of mood*. Unpublished manuscript. Retrieved from https://stimson.web.unc.edu/wp-content/uploads/sites/9919/2024/01/Moodbio.pdf

van Atteveldt, W., van der Velden, M. A. C. G., & Boukes, M. (2021). The validity of sentiment analysis: Comparing manual annotation, crowd-coding, dictionary approaches, and machine learning algorithms. *Communication Methods and Measures*, 15, 121–140. doi:10.1080/19312458.2020.1869198

Vick, K. (2022, November 7). Americans say they feel pretty good—At least compared to the country as a whole. *TIME*. Retrieved from https://time.com/6230100/america-national-mood-poll/

Wlezien, C. (2024). News and public opinion: Which comes first?. *The Journal of Politics*, 86, 1–17. doi:10.1086/726940

Wlezien, C., & Erikson, R. S. (2001). Campaign effects in theory and practice. *American Politics Research*, 29, 419–436. doi:10.1177/1532673X01029005001

Wlezien, C., & Soroka, S. (2024). Media reflect! Policy, the public, and the news. *The American Political Science Review*, 118, 1563–1569. doi:10.1017/S0003055423000874

Wlezien, C., Soroka, S., & Stecula, D. (2017). A cross-national analysis of the causes and consequences of economic news. *Social Science Quarterly*, 98, 1010–1025. doi:10.1111/ssqu.12445

York, C. (2013). Cultivating political incivility. *Electronic News*, 7, 107–125. doi:10.1177/1931243113507926

Young, L., & Soroka, S. (2012). Affective news: The automated coding of sentiment in political texts. *Political Communication*, 29, 205–231. doi:10.1080/10584609.2012.671234

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